

- 2 (a) (i) $v = u + at$ C1
 $= 4.23 + 9.81 \times 1.51$ M1
 $= 19.0(4) \text{ ms}^{-1}$ (Allow 2 s.f.) A0 [2]
 (Use of $-g$ max 1/2. Use of $g = 10$ max 1/2. Allow use of 9.8. Allow 19 ms^{-1})
- (ii) either $s = ut + \frac{1}{2} at^2$ (or $v^2 = u^2 + 2as$ etc.)
 $= 4.23 \times 1.51 + 0.5 \times 9.81 \times (1.51)^2$ C1
 $= 17.6 \text{ m}$ (or 17.5 m) A1 [2]
 (Use of $-g$ here wrong physics (0/2))
- (b) (i) $F = \Delta P / \Delta t$ need idea of change in momentum C1
 $= [0.0465 \times (18.6 + 19)] / 12.5 \times 10^{-3}$ C1
 $= 140 \text{ N}$ A1
 (Use of $-$ sign max 2/4. Ignore $-ve$ sign in answer)
 Direction: upwards B1 [4]
- (ii) $h = \frac{1}{2} \times (18.6)^2 / 9.81$ C1
 $= 17.6 \text{ m}$ (2 s.f. -1) A1 [2]
 (Use of 19 ms^{-1} , 0/2 wrong physics)
- (c) either kinetic energy of the ball is not conserved on impact
 or speed before impact is not equal to speed after hence inelastic B1 [1]

- 3 (a) Resultant force (and resultant torque) is zero B1